

Derek Wang

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EDUCATION

University of Waterloo

Expected April 2030

Bachelor of Applied Science in Computer Engineering (Co-op)

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Prandtl Dynamics Inc.

Toronto, ON

- Tracked 5+ moving targets at 170 m with 20 ms latency by chaining SRP-PHAT azimuth/elevation estimation, delay-and-sum beamforming per source, and Unscented Kalman Filter + Global Nearest Neighbor tracking
- Raised acoustic drone detection to 98% mAP with 40% fewer false alarms and half the model size via EfficientAT
- Closed cross-tenant and auth gaps across 6 dashboard services with middleware that injects org-ID predicates into every query, scopes WebSocket rooms per org, and fails closed on JWT checks, enforced by CI audit gates
- Automated cleaning of 50+ hours of training audio via a Python tool localizing and beamforming array recordings
- Slashed recording setup from 10 to 2 minutes by swapping a tethered rig for a Raspberry Pi audio service over SSH

Software Engineer Intern

March 2026 – June 2026

Kenoxis Technologies Inc.

Mississauga, ON

- Lowered API latency by 35% by caching posting lookups and auth sessions in Redis across Node.js microservices
- Designed MongoDB schemas and ingestion pipelines for job postings, improving refresh from daily to hourly
- Flagged suspicious applicants 60% faster by queuing LLM document review for fraud, scoring, and LMIA results
- Onboarded 5+ interns with internal API docs and integration guides, cutting repeat API questions to the team

Autonomy Software Developer

September 2025 – Present

WATonomous – Student Design Team

Waterloo, ON

- Built a ROS 2 navigation stack running A* path planning over a 100x100 m costmap and Pure Pursuit control under 10 ms per cycle, replanning at 10 Hz for real-time autonomous rover navigation
- Led YOLO11s model retraining, improving detection mean average precision from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Accelerated YOLO11s inference to 30 FPS on a Jetson Nano via ONNX export and onnxslim graph simplification

PROJECTS

7-DOF Robotic Arm Motion Control | C++17, Eigen, MuJoCo, CMake, React, TypeScript

- Derived Lie-group kinematics and recursive Newton-Euler dynamics in C++17/Eigen, matching MuJoCo to 1e-13
- Held sub-0.2 mm tip error with zero actuator saturation over 50+ scenarios via jerk-limited 1 kHz computed torque
- Cut worst-case IK replan latency from 40.7 to 10.5 ms across 324 targets via damped least squares with 26 seeds
- Resolved redundancy via annealed null-space projection of 4 objectives and swept joint-space collision checks

Autonomous Vehicle State Estimation & Tracking | C++, Eigen, ROS 2, Python, Docker, KITTI

- Implemented a 15-state error-state Kalman filter fusing 100 Hz IMU with 10 Hz GPS in C++/Eigen on ROS 2
- Localized the ego vehicle to 0.558 m mean error over 8 KITTI sequences (6.6 km), cross-checked via evo
- Reached 0.886 MOTA on ground-truth detections with a CV/CA/CT IMM of UKFs and Hungarian association
- Coupled tracking to localization by world-framing detections with the filter's pose (error correlation 0.976)

Reparo - AI Repair Tool (Hack Canada 2026 Winner) | Python, FastAPI, Swift, Gemini

- Turned a photo of a broken device into a structured repair plan (steps, cost, difficulty, repairability score) in under 60 s by chaining Gemini vision for failed-component detection with Gemini reasoning and schema-validated parsing
- Sourced replacement parts automatically by chaining SerpAPI search, Shopify storefront discovery, and Shopify JSON endpoints, mapping Gemini's parts list to a purchasable cart inside a no-install Swift App Clip

TECHNICAL SKILLS

Languages: C++, Python, MATLAB, Java, JavaScript, TypeScript, SQL

Frameworks/Libraries: ROS 2, PyTorch, OpenCV, Eigen, MuJoCo, ONNX, React, Next.js, Node.js, FastAPI

Infrastructure: Docker, GitHub Actions, CMake, Git, SQLite, MongoDB, Redis, Foxglove